The Bitcoin OTC Trust Network: Structure and Rating Patterns

Ouko Oyombe[1*], Sheillah Khaluvitsi[1] and Abdihakim Yussuf[1]

[1*]Network Science, Eötvös Loránd University, Budapest, Hungary.

Abstract

Online trading platforms use reputation systems to help users assess unfamiliar traders. This study examines the structure and development of the Bitcoin OTC web of trust, using 35,592 signed, directed ratings among 5,881 accounts. Positive ratings are analyzed as a network and compared with random, Barabási–Albert (BA), and degree-preserving reference networks. The network has prominent hubs, more clustering than all three references, and a degree tail heavier than BA predicts, although lognormal and cut-off tails fit it better than a pure power law. Accumulated positive ratings became more concentrated: the top 1% of users held 15.5% at the end of 2011 and 23.5% at the end of 2015, and the rise holds within a fixed 2011 cohort. The apparent post-2013 fall in the concentration of each year's new ratings is not supported at equal volume. Users with at least 32 earlier positive ratings received new ratings 12.7 times as often as expected under uniform choice, but only 1.65 times as often as users with similar observed tenure and recent activity. Simulated removal of highly connected users fragmented the network far faster than random removal. These findings describe recorded ratings; they do not establish why users received ratings, whether trades took place, or whether users were trustworthy.

Keywords: network science, signed networks, trust ratings, network evolution, Bitcoin OTC

1 Introduction

People trading through an online platform must often assess others with limited information. Bitcoin OTC addressed this problem through a reputation system in which members could give positive or negative ratings to other members [1]. The resulting network records whose opinions were directed toward whom. Its structure can reveal

Table 1 Bitcoin OTC rating records and graph counts.

Quantity	Value
Rating records / unique directed pairs	35,592
Users appearing as source or target	5,881
Positive / negative / zero ratings	32,029 / 3,563 / 0
Self-ratings / repeated directed pairs / missing cells	0 / 0 / 0
Undirected links after combining reciprocal ratings	18,591
Reciprocal positive pairs combined into single links	13,438

whether ratings are distributed widely or concentrated around a small number of users, and whether those users occupy prominent positions in the network. A rating is an opinion, however, and does not establish that a transaction occurred or that the recipient was trustworthy.

The dataset provides the sign, strength, direction, and time of each recorded rating. Kumar et al. [2] studied weighted signed networks, including Bitcoin OTC, to predict the strength of a rating. Here we use the records for a different question: how the network's positive-rating structure compares with network models and their theoretical expectations, and how the distribution of incoming ratings changed over time. A prominent user may have received many ratings for several reasons, including activity, tenure, or visibility; a degree distribution alone cannot distinguish these explanations.

Our focused question is: *How did positive ratings become concentrated among Bitcoin OTC users, and how sensitive is the apparent advantage of already highly rated users to the comparison benchmark?* Structural analysis establishes the setting: we define the rating graphs, measure their connectivity and communities, and compare an undirected representation with Erdős–Rényi (ER), Barabási–Albert (BA), and degree-preserving references [3, 4]. The temporal investigation distinguishes accumulated history from ratings issued within each year, then compares subsequent recipient choices with several benchmarks. These analyses are exploratory rather than preregistered tests. A high-degree tail cannot alone demonstrate preferential attachment; timestamps permit a more direct, though still observational, test of target selection [5, 6].

2 Data and network construction

2.1 Source and meaning

We use the Bitcoin OTC web of trust dataset distributed by the Stanford Network Analysis Project (SNAP) [1, 2, 7]. It contains 35,592 ratings among 5,881 user accounts, recorded from 8 November 2010 to 25 January 2016. Each record identifies the member giving the rating, the member receiving it, the score, and the time. Scores range from -10 (strong distrust) to $+10$ (strong trust). The data therefore form a directed, signed, and weighted network with a time for every link.

The first and last calendar years are incomplete. A *node* is a user account and an *edge* (or link) is a rating from one account to another. The arrow points from the rater to the recipient; it does not represent a Bitcoin transfer. The dataset contains no

repeated records for the same ordered pair, so it does not allow us to trace revisions to a rating. An absent edge means no rating is recorded; it does not establish that the two users never interacted.

We retain each distinct integer ID as a separate account because the dataset provides no information for merging identities. The audit (Table 1) found nothing to remove, and no sampling was needed. For example, user 6 gave user 2 a +4 rating in 2010, and user 2 gave user 6 a +5 rating in 2012. In the directed network these are two distinct ratings; in the undirected network of positive ratings they become one link $\{2, 6\}$. The repository documents the reconstruction and the source checksum.

2.2 Directed and undirected networks

The main analysis keeps positive ratings, meaning scores greater than zero. In this *directed network*, each link points from rater to recipient. It contains all 5,881 observed accounts and 32,029 positive directed links. The 308 accounts without any positive link remain as isolated nodes, because dropping them would silently change every denominator. In-degree k_i^{in} counts positive ratings received by account i; out-degree k_i^{out} counts positive ratings given. Both average $\langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle = L/N = 5.45$. These degrees count links: a +2 and a +9 each add one to the recipient's in-degree. The score remains an attribute of each directed link but is not used as a weight when calculating the reported network measures. Scores are used to select positive ratings and to repeat the temporal analysis for ratings of at least +5. The Cytoscape static view uses the original signed scores to show rating sign and strength.

For methods that require an *undirected network*, we ignore the direction of each positive link. Two accounts are connected if either one rated the other positively. Combining reciprocal ratings gives 18,591 undirected links among the same 5,881 accounts. This representation loses the rating direction and does not use the scores as weights, so results on it describe topology only. We use it for model comparisons, communities, clustering, components, and node removal. The directed network is used for in-degree, out-degree, directed betweenness, and the temporal recipient analysis. Negative ratings remain in the full dataset and Cytoscape view, but are excluded from these two networks of positive ratings.

3 Methods

3.1 Structural measures

For the undirected network, N is the number of accounts and L the number of links. The degree k_i of account i is its number of linked neighbors, so $\langle k \rangle = 2L/N = 6.322$. Density $2L/[N(N-1)]$ is the fraction of possible account pairs that are linked, here 0.001075.

We report two clustering measures: *transitivity*, three times the number of triangles divided by the number of connected triples, and average local clustering, the mean of $C_i = 2e_i/[k_i(k_i-1)]$ with $C_i = 0$ when $k_i < 2$. Whole-network mean path length and diameter are undefined because the undirected network is disconnected, so we

calculate both exactly on the largest component. Reference graphs use 32 fixed-seed source nodes for mean path length.

All reported distances are unweighted hop counts: each link in the undirected network has length one. We do not use the score as path length, because a stronger favorable opinion is not a longer distance, and an inverse-score cost would require an independently justified trust-propagation model. Paths through ratings should be interpreted as graph proximity, not trade or trust routes.

We compute positive directed in- and out-degree and exact normalized betweenness on the directed network of positive ratings, and compare the exact values with an estimate from 64 fixed-seed source nodes. Betweenness describes positions on shortest paths; those paths need not carry trust or money. Louvain modularity optimization identifies communities in the undirected network [8, 9]. For the five largest groups, we calculate sizes and the total internal degree $2L_{\text{in}}$ and external degree L_{out} of each group; a group whose internal share $2L_{\text{in}}/(2L_{\text{in}} + L_{\text{out}})$ exceeds one half is a *weak community* [6, Ch. 9]. We rerun Louvain with 20 seeds to assess stability and on reference graphs to compare modularity. The groups are algorithmic partitions without independent social labels. NetworkX supplies the graph computations [10].

3.2 Reference models and theoretical expectations

We generate ten ER networks $G(N, L)$ with $N = 5{,}881$ accounts and exactly $L = 18{,}591$ links, and ten BA networks with three links per new node, giving 17,634 links, 5.15% fewer than observed. Both use seeds 2026–2035. Their means and standard deviations describe variation between simulated networks, not uncertainty on the observed network. We also make eight degree-preserving randomizations ("rewires"), each swapping the ends of two links ten times per link [11], seeds 2026–2033. These keep every user's degree fixed but not direction, sign, or community structure. We measure degree correlations with the assortativity coefficient r [12] and the degree correlation function $k_{nn}(k)$, the mean degree of the neighbors of degree-k users, in the undirected network and in the four in-/out-degree combinations of the directed network, and compare each with the same eight seeds (directed rewires preserve both in- and out-degree).

We set simulations beside analytical expectations [6]. For ER these are $\sigma_k = \sqrt{\langle k \rangle}$, a largest degree from $N\,P(K \geq k_{\max}) = 1$, a giant component from $S = 1 - e^{-\langle k \rangle S}$, $\langle d \rangle \approx \ln N / \ln \langle k \rangle$, and $\langle C \rangle = p$. For BA they are $\gamma = 3$, $k_{\max} \approx m\sqrt{N}$, $\langle d \rangle \approx \ln N / \ln \ln N$, and $\langle C \rangle \sim (\ln N)^2 / N$. These are large-network approximations.

A straight log–log plot cannot establish a power law [13]. We therefore estimate γ by maximum likelihood, choose $k_{\min}$ where the fitted and observed tails differ least, bootstrap the uncertainty (200 resamples), test whether the data could come from the fitted power law (500 synthetic samples), and compare it with lognormal, exponential, and cut-off tails [13, 14]. We calibrate the estimator on simulated graphs with known exponents.

BA adds every link with a new node, so we classify each undirected link when it first appears, giving m links per new user and n links between existing users per new user. A user counts as existing from their first positive rating, so a newcomer's later links count as internal. Barabási's internal-link extension of BA adds, in each time step, a new node with m links followed by n internal links, each joining two

existing nodes chosen in proportion to their degrees (double preferential attachment). Its degree exponent is [6, Sect. 6.5, Eq. 6.27]

$$\gamma = 2 + \frac{m}{m + 2n}, \tag{1}$$

which lies between 2 and 3 and returns BA's $\gamma = 3$ when $n = 0$. If internal links instead joined randomly chosen pairs, the exponent would be $\gamma = 3 + 2n/m \geq 3$ [6, Eq. 6.28]. We also simulate the double preferential extension with the measured rates.

3.3 Node-removal experiment

We remove 0.1%, 0.5%, 1%, 2%, 5%, and 10% of users from the undirected network, either as a targeted attack in descending order of *original* degree (ties by user ID) or as random failures in 30 seeded replicates, and measure the largest connected component as a share of the original 5,881 users. Across the full range of removal fractions (20 replicates, one realization of each reference graph), we compare the results with the random-graph benchmark $f_c = 1 - 1/(\kappa - 1)$ with $\kappa = \langle k^2 \rangle / \langle k \rangle$, the random-failure threshold of an infinite uncorrelated network with the same degrees, from the Molloy–Reed criterion $\kappa > 2$ [15, 16]. This is a hypothetical graph intervention, not an observed history of user departures, and fixed original degree differs from an adaptive attack.

3.4 Focused temporal investigation

We order ratings by their timestamps and build a snapshot at the end of each year from 2011 through 2015. Each snapshot contains all positive ratings recorded up to that date, and its nodes are users who have appeared in a positive rating by that date, including those with zero incoming positive degree. We calculate the share of incoming positive ratings received by the highest-rated 1% of these users, rounding the number of users up, and the Gini coefficient of incoming positive ratings (0 when every user receives the same number, near 1 when one user receives almost all). The snapshots show growth in the recorded history; they do not establish whether earlier ratings remained current. We repeat the analysis for ratings of at least +5.

Three further checks distinguish accumulation from changes in population and volume. First, annual-only graphs retain ratings issued within each calendar year and all users giving or receiving them. Second, a fixed-cohort calculation follows the 1,631 users observed by the end of 2011, counting all cumulative positive ratings they receive, including those from later entrants. Third, because the size of the top 1% group and rating volume both differ by year, we redraw every year 1,000 times to 743 ratings (80% of the smallest year) and record the share of the four largest recipients, reselected in each sample.

The event-level comparison uses the chronological positive ratings. Before a rating $u \to v$, eligible candidates are users previously observed in a positive rating, excluding the rater u and anyone u has already rated positively. If v has not appeared before, the event is excluded. We group candidates by how many positive ratings they had

already received and compute

$$R_k = \frac{O_k}{E_k} \tag{2}$$

where O_k is the number of ratings whose recipient was in group k immediately before the rating, and E_k adds, over eligible events, the fraction of candidates in group k. A ratio of one matches the benchmark. Groups are 0, 1, 2–3, 4–7, 8–15, 16–31, and 32 or more previous positive ratings. The source has no tied positive-rating timestamps, so event ordering is unambiguous.

We repeat the comparison within the chosen target's *prior observed tenure and activity group*. Tenure is days since first appearance in a positive rating (under 90, 90–364, or at least 365 days). Activity is the number of positive ratings given or received during the preceding 90 days (0, 1–2, 3–9, or at least 10). We require at least five candidates. To separate the factors, we also use tenure only, activity only, tenure with ratings *given* only (which removes the overlap between activity and received ratings), and only the first rating in each pair. For each benchmark we fit the nonlinear preferential attachment rule $\Pi(k) \propto (k+1)^{\alpha}$ [6] by maximum likelihood over the candidate sets [17]: $\alpha = 1$ is BA's linear rule and $\alpha < 1$ is sublinear. Standard errors treat events as independent, so they are optimistic.

4 Results

4.1 Connectivity and heterogeneity

The undirected network of positive ratings is sparse: only about 0.11% of possible user pairs have a link, and average degree is 6.322. Its largest component contains 5,551 users, or 94.39% of all 5,881 observed accounts. However, 308 of its 317 components are isolates created by retaining accounts without any positive link, and among the 5,573 users with a positive link, 99.61% belong to the largest component. The component's diameter is 14 and its exact mean shortest path is 3.761 links; the 32-source estimate of 3.559 was 5.4% too low. Degrees vary substantially: the standard deviation is 20.4, the 95th percentile is 23, and the maximum is 788 (Fig. 1). Transitivity is 0.0570, while average local clustering is 0.155 (0.272 among users with degree at least 2). Transitivity is dominated by hubs, whose many neighbor pairs rarely close, whereas the local average weights each user equally.

Figure 2 shows the undirected network of positive ratings using a seeded spring layout, with disconnected pieces in a separate panel. Positions aid inspection and do not measure geographic, temporal or social distance.

4.2 Comparison with theoretical expectations

The ER and BA simulations agree with their formulas, which supports the machinery: $N\,P(K \geq k_{\max}) = 1$ gives 17 against a simulated 17.1, and $m\sqrt{N}$ gives 230 against 231. The observed network departs from both (Table 2). Its largest hub is 46 times the ER maximum and 3.4 times the BA maximum, $\langle k^2 \rangle / \langle k \rangle$ is 71.9 against BA's 17.4, and local clustering is 154 times ER's. Mean distance (3.761) is shorter than ER's (4.91), which together with high clustering is the small-world property [18]. The rewires,

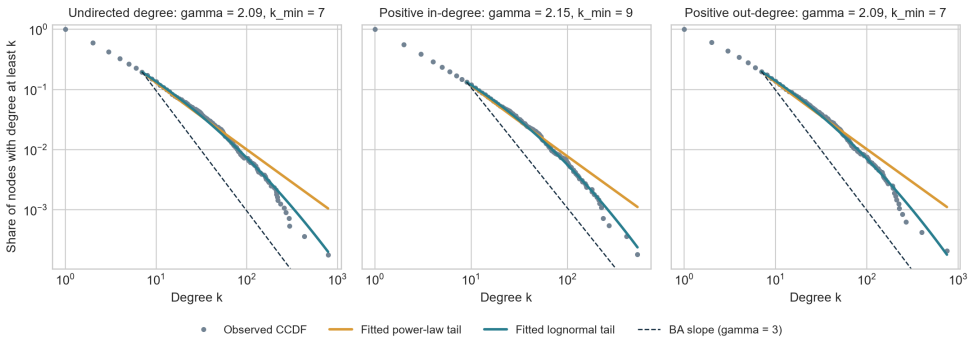


Fig. 1 Complementary cumulative degree distributions with fitted tails. Above about $k = 100$ the data bend below the fitted power law, and all three tails are shallower than BA's slope.

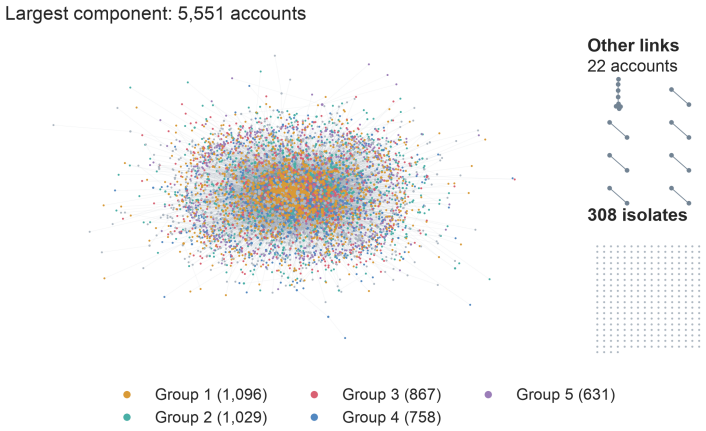


Fig. 2 Undirected network of positive ratings. The main panel shows the 5,551-user largest component; the right panel shows the remaining 22 linked users and 308 isolates. Colors indicate the five largest Louvain groups and size reflects degree. Direction, sign and score are omitted.

which keep every degree, come much closer. Rewired transitivity is 0.0430 ± 0.0011 (range 0.0407–0.0440) against the observed 0.0570, and observed local clustering is 2.5 times the rewired value, so hubs explain the short paths and most of the transitivity, but less than half of the local clustering. The network is disassortative ($r = -0.145$) [12], yet the rewires already reach -0.114, so most of this correlation follows from the degree sequence: the largest degree, 788, is about four times the structural cutoff $k_s = \sqrt{\langle k \rangle N} \approx 193$, above which a simple graph cannot give hubs enough high-degree partners [6, Ch. 7]. The remaining -0.030 lies outside all eight rewires (-0.118 to -0.112), and each directed variant lies 0.034–0.037 below its rewired mean. The degree correlation function falls as $k_{nn}(k) \propto k^{\mu}$ with $\mu = -0.23$, against -0.10 for the rewires, and the 2,226 single-link users have neighbors with 181 links on average against 93 after rewiring. Hub-to-periphery wiring thus exceeds what the degrees impose, though by a small margin.

Table 2 Observed undirected network against formulas and simulations (means of ten ER, ten BA, eight rewired, and ten internal-link graphs). LCC is the largest connected component. The observed mean path is exact; simulated paths use 32 sources. Fitted γ under BA and internal-link simulations is the estimator applied to those graphs. Internal-link graphs contain only the 5,573 users with positive links, so they have no isolates.

Quantity	Observed	ER formula	ER sim.	BA formula	BA sim.	Rewired	Int. link
Mean degree $\langle k \rangle$	6.32	6.32	6.32	6.00	6.00	6.32	6.68
Degree std. dev.	20.4	2.5	2.5	–	8.3	20.4	23.1
Maximum degree	788	17	17	230	231	788	333
$\langle k^2 \rangle / \langle k \rangle$	71.9	7.3	7.3	–	17.4	71.9	86.4
LCC share	0.944	0.998	0.998	1.000	1.000	0.927	1.000
Mean path (LCC)	3.76	4.71	4.91	4.02	4.08	3.52	3.48
Avg. local clustering	0.1550	0.0011	0.0010	0.0128	0.0081	0.0615	0.0974
Transitivity	0.0570	0.0011	0.0011	–	0.0042	0.0430	0.1164
Assortativity	−0.145	0.000	0.001	–	−0.044	−0.114	−0.142
Modularity	0.504	–	0.369	–	0.388	0.334	0.324
Fitted γ	2.09	–	–	3.00	2.74	–	1.88

Is the tail a power law? Fitted exponents are 2.09 for undirected degree (95% bootstrap interval 1.91–2.24, $k_{\min} = 7$), 2.15 for in-degree (1.86–2.28, $k_{\min} = 9$), and 2.09 for out-degree (1.93–2.78, $k_{\min} = 7$). The goodness-of-fit test rejects the pure power law in all three cases ($p < 0.002$: no synthetic sample fits as badly). It beats an exponential tail decisively, but a lognormal ($p \leq 0.008$) and a power law with an exponential cut-off ($p < 0.001$) fit significantly better. Synthetic tails are drawn exactly from the discrete power law, but the tests treat degrees as independent, so these p-values and intervals are approximate. The fitted law would extrapolate to a largest degree near 4,290, far above the observed 788, matching the bend in Fig. 1, and $k_{\min}$ is weakly identified (bootstrap interval 3–11). The defensible claim is that the network is strongly heterogeneous and hub-dominated with a tail heavier than BA's, not that it is strictly scale-free.

Internal links. Of the 18,591 links, 13,100 (70.5%) joined two users who had already appeared. Because this includes a newcomer's later links, it describes the records rather than refuting BA's growth rule. The measured rates are $m = 0.99$ and $n = 2.35$, so Eq. (1) predicts $\gamma = 2.17$, inside the observed interval, whereas BA's 3 and the $\gamma \geq 3$ of random internal links lie outside it. The estimator runs low on finite graphs, however: it returns 2.74 on BA graphs (true value 3) and 1.88 on the simulated extension, so on that common scale the observed 2.09 lies between the two and closer to the extension. Internal links therefore matter without fully accounting for the tail. The extension also moves $\langle k^2 \rangle / \langle k \rangle$ (86.4), assortativity (−0.142), and mean path (3.48) close to the data, but its largest hub (333) stays well below 788, its transitivity (0.116) overshoots, its local clustering (0.097) falls short of 0.155, and its modularity (0.324) is far below 0.504. Degree-driven growth does not produce the observed community structure.

4.3 Centrality, communities, and Cytoscape view

The highest positive in-degree in the directed graph is 535 (anonymous user ID 35), followed by 411 (ID 2642) and 270 (ID 1810). These values count positive ratings received; they do not measure verified reliability. User 35 also has the highest exact

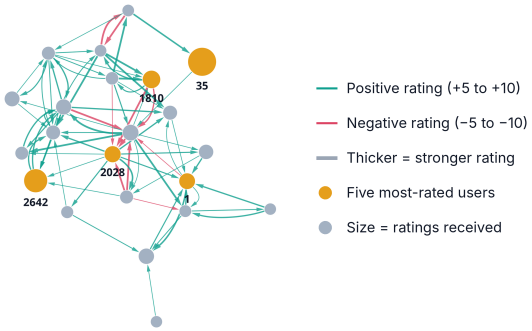


Fig. 3 Cytoscape spotlight of strong ratings ($|\text{score}| \geq 5$) among the 25 users with highest positive in-degree in the full data; three users isolated by that filter are omitted, leaving 22 users and 76 directed edges. Selection uses future information and is for illustration, not a representative view.

normalized betweenness, 0.1425, followed by 2642 (0.0616), 1 (0.0440), and 7 (0.0400). In-degree and out-degree move together (Spearman 0.84), as high reciprocity suggests, and betweenness largely follows in-degree (Spearman 0.87; nine of the top ten overlap), so hubs are also the main structural bridges. User 1810 is an exception: third by in-degree but seventh by betweenness, it bridges fewer shortest paths than its rating count suggests. The 64-source estimate is adequate for overall ranks (Spearman 0.95) but noisy for individuals: it gave user 7 a value of 0.0628, and only 13 of its top 20 match the exact top 20. High betweenness does not measure actual transactions or information flow.

Louvain returns modularity 0.504 and 341 groups, of which 308 are singleton isolates and 33 contain at least two users. The five largest contain 1,096, 1,029, 867, 758, and 631 users, with internal shares of 68.4%, 75.1%, 63.1%, 55.8%, and 64.2%, so all five are weak communities. Louvain optimizes a related criterion, so these profiles describe its partition rather than independently validate communities or identify trading roles. Across 20 seeds, modularity stays between 0.487 and 0.517, yet the largest group ranges from 1,024 to 1,457 users, and other partitions agree only moderately with the reported one (mean normalized mutual information 0.66). The same procedure gives lower modularity on rewired (0.334), ER (0.369), and BA (0.388) graphs. Community organization is therefore a robust feature of the network, while the membership of any particular group is not.

Figure 3, prepared in Cytoscape [19], restores the direction and sign omitted from the undirected representation. A companion CyAnimator video [20] connects five year-end states from 2011 to 2015, showing 5, 16, 54, 73, and 76 recorded strong ratings. Interpolated fades are visual transitions, not additional observations.

4.4 Robustness to node removal

The largest component initially contains 94.4% of users. Removing the highest-degree 1% reduces that share to 72.0%, compared with 93.1% under random failures. At 5% removed, the targeted result is 45.6% versus 87.6%; at 10%, it is 14.3% versus 81.2%. The benchmark f_c is 0.986 for the observed degrees ($\kappa = 71.9$), 0.934 for BA, and

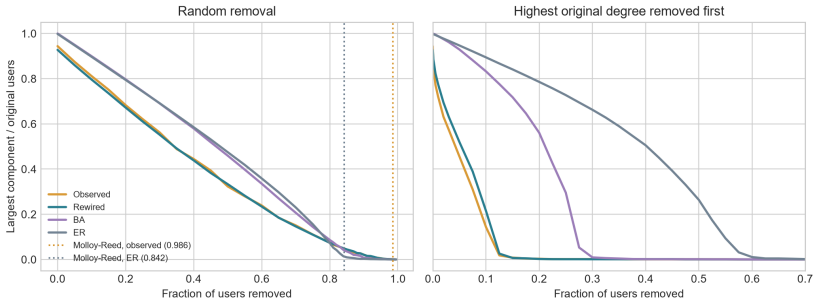


Fig. 4 Largest component as a fraction of the original users after random or targeted removal, with one realization of each reference graph. Dotted lines mark the random-graph benchmarks f_c. The observed curve starts at 0.944 because 308 isolates remain in the denominator.

0.842 for ER, and simulations agree on the ordering (Fig. 4). The largest component falls below 1% of users at $f = 0.85$ for ER, close to its benchmark, and at $f = 0.93$ for the observed network. The two differ in kind: the simulation uses this finite, clustered network and a 1% cutoff. A targeted attack reverses the picture, as theory predicts for heterogeneous networks [21]: the observed network collapses at $f = 0.15$, against 0.30 for BA and 0.625 for ER. The rewired network collapses at the same fraction (0.15), so this fragility follows mainly from the degree sequence. This does not predict how trust or trading would change if real accounts disappeared.

4.5 Evolution of positive rating concentration

The recorded positive-rating network grew from 1,631 observed users and 7,745 cumulative positive ratings at the end of 2011 to 5,571 users and 31,989 ratings at the end of 2015. During this period, the top 1% share of cumulative incoming ratings rose from 15.5% to 23.5%, and in-degree Gini rose from 0.611 to 0.691 (Table 3). This describes increasingly unequal accumulated counts, not current reputation. The result persists within a fixed population: among the same 1,631 users, the share rises from 15.5% to 22.2%. With ratings of at least +5, the sample is much smaller (1,585 users and 2,885 ratings in 2015), but the share still rises from 0.163 to 0.193 and Gini from 0.550 to 0.669.

The raw annual-only share of ratings issued within each year is 15.3%, 19.7%, 21.0%, 14.4%, and 11.3% from 2011 to 2015, which appears to peak in 2013 and then fall (Fig. 5). These years are not directly comparable, however: the 2013 top group has 25 users sharing 11,138 ratings, while the 2015 group has 4 users sharing 929. When every year is redrawn to 743 ratings, the 2013 share is 12.1% (95% range 10.1–14.1) and the 2015 share is 10.0% (8.3–12.2), with overlapping ranges, and the four largest recipients in each sample hold 7.8% and 11.6%. The raw fall is thus sensitive to volume and to the size of the top group. As redrawing also changes which users appear, this does not show that volume alone caused the fall, only that the data do not support a claim that new ratings became clearly less concentrated after 2013.

The network's shape also changes. Among users observed in positive ratings, the largest component stays above 99.6% at each year end. Transitivity of the undirected

Table 3 Cumulative snapshots and annual-only concentration. Each top 1% is recalculated within its own observed population, including zero-incoming users. The last column redraws each year to 743 ratings. Partial years 2010 and 2016 are not displayed; cumulative counts include 2010.

Year	Cum. users	Cum. ratings	Cum. top 1%	Cum. Gini	Annual top 1%	Equal vol.
2011	1,631	7,745	0.155	0.611	0.153	0.095
2012	3,116	16,367	0.199	0.659	0.197	0.112
2013	4,991	27,505	0.225	0.679	0.210	0.121
2014	5,455	31,060	0.232	0.688	0.144	0.096
2015	5,571	31,989	0.235	0.691	0.113	0.100

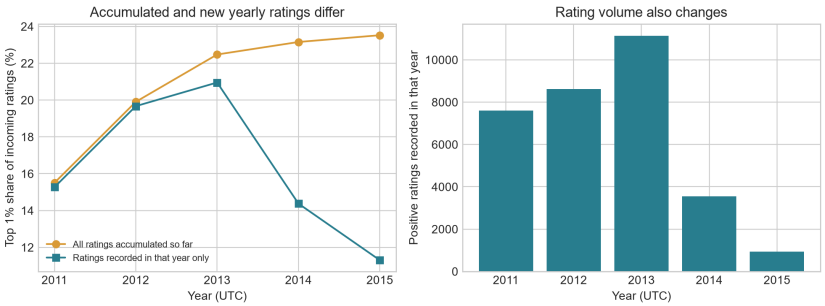


Fig. 5 Cumulative and raw annual-only incoming-rating concentration, with rating volume on the right. The raw fall after 2013 coincides with the fall in volume; Table 3 shows that it shrinks at equal volume.

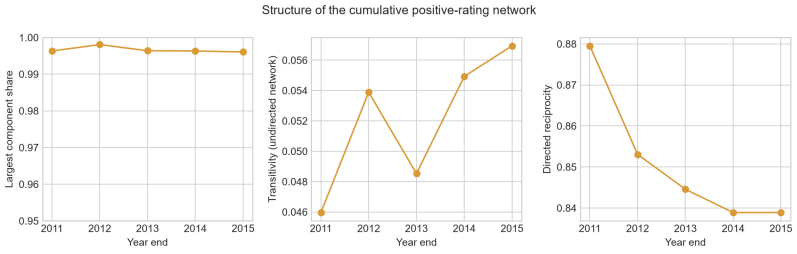


Fig. 6 Largest-component share, transitivity of the undirected network, and directed reciprocity at UTC year ends in the cumulative positive-rating network.

network moves from 0.0460 to 0.0569, with a dip in 2013, and directed reciprocity falls from 0.8795 to 0.8389 (Fig. 6). These are changes in the cumulative record, not evidence that existing relationships persisted.

4.6 Who receives the next rating?

Of 32,029 positive rating events, 27,133 have a target already observed and an eligible candidate set; the other 4,896 first-time-target events are excluded. Under uniform choice, the 32+ prior in-degree group has 7,356 observed events versus 579.4 expected

Table 4 Observed/expected recipients by prior positive in-degree under each benchmark, with the attachment exponent α and its standard error.

Benchmark	Events	R_0	R_{2-3}	R_{32+}	α (s.e.)
Uniform	27,133	1.76	0.48	12.70	1.07 (0.005)
Tenure only	27,133	1.25	0.49	18.46	1.39 (0.005)
Activity only	27,105	3.94	0.77	1.71	0.36 (0.006)
Tenure and activity	26,993	3.85	0.77	1.65	0.68 (0.009)
Tenure and ratings given	26,588	2.75	0.85	1.44	0.61 (0.009)
Uniform, first rating in a pair	13,695	0.72	0.55	11.09	1.00 (0.007)
Tenure and activity, first rating in a pair	13,617	1.16	0.86	1.63	0.75 (0.013)

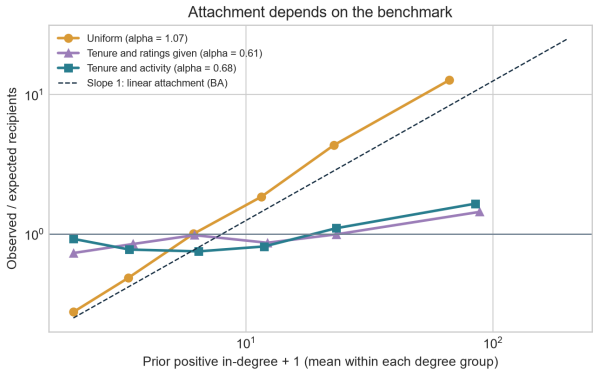


Fig. 7 Observed/expected recipients against prior in-degree under three benchmarks. Matching on recent activity flattens the near-linear relationship seen under uniform choice. This is a descriptive benchmark, not a causal effect estimate.

($R_k = 12.70$), the 16–31 group has 4,001 versus 922.9 ($R_k = 4.34$), and the degree-one group is underrepresented ($R_k = 0.28$). The 32+ ratio exceeds 11 in each complete year, and the fitted exponent is $\alpha = 1.07$, close to the linear attachment BA assumes (Table 4, Fig. 7). At first sight this supports preferential attachment.

The result does not survive the benchmark. Matching on tenure and activity retains 26,993 events (99.48%) and reduces the 32+ ratio to 1.65 and α to 0.68, a sublinear rule. Separately, tenure alone raises the ratio to 18.46, while recent activity alone reduces it to 1.71. Because activity includes ratings received, the matching might absorb the effect it tests; counting only ratings the candidate *gave* still gives 1.44, so the reduction persists without that overlap. Most of the uniform contrast is associated with how recently and how often a candidate itself rated others. A residual remains under every benchmark.

Reciprocity also matters. Of the eligible events, 13,438 (49.5%) return an earlier rating from the recipient, with a median delay of 3.7 minutes and 87.3% within a week. These look like mutual feedback after an interaction rather than independent choices. They explain the elevated matched ratio for users with no prior ratings (3.85): 79% of those events are returns, and the ratio falls to 1.16 when only the first rating in each

pair is counted. The 32+ result barely moves (1.63), so returns do not account for it in this check.

5 Discussion

Theory against observation. ER provides only a baseline. BA captures hub formation and, under a uniform benchmark, the roughly linear attachment it assumes, but it understates hubs and misses clustering. The internal-link extension moves the tail, heterogeneity, and degree correlations toward the data, but still understates the largest hub and community structure. The benchmark f_c correctly ranks tolerance to random failures, and the prediction of robustness to random failures with fragility to targeted attacks holds. A degree-only mechanism fits well only under the weakest benchmark, so agreement with BA does not establish it as the generating process. Component sizes deserve less weight because the positive filter creates isolates absent from ER and BA.

Implications for the real system. Reputation is concentrated and stays so: by 2015, 56 accounts held 23.5% of positive ratings. Old ratings stay in this record while new ones fell sharply after 2013, so a large accumulated score may increasingly describe past standing. In simulation, the rating network also depends on few accounts: removing about 1% of prominent accounts disconnects far more of it than removing the same number at random. A rating count should not be read as independent endorsements, because about half of eligible ratings return an earlier one, usually within minutes. Finally, recent participation may do more for attracting ratings than an established reputation alone, though it cannot be separated from unobserved trading volume.

Limitations and sensitivity. Ratings are opinions, not verified transactions or fraud outcomes. The source has no registration dates, exposure, trade volume, rating deletions, or full membership list, and anonymous IDs may not match people one to one, so tenure means time since the first positive record. The undirected representation collapses 13,438 reciprocal positive pairs and omits signs, so model and community results apply to it only. Tenure and activity groups are coarse, percolation curves use one realization of each reference graph, and formula values are asymptotic. Stable across variants: rising accumulated concentration, the high-degree residual, and hub fragility. Not stable: the high-degree ratio (1.44 to 18.46 across benchmarks), the yearly trend (sensitive to volume), and specific Louvain groups across seeds.

Further work. A joint choice model could estimate degree, recency, and reciprocity effects together and test them out of sample, and could add fitness and aging terms. Time-respecting null models would strengthen the temporal comparisons. Signed triad analysis could use the negative ratings set aside here, adaptive attacks could recompute degree after each removal, consensus clustering could stabilize the communities, and external fraud labels could test whether network position relates to trustworthiness.

6 Conclusion

The undirected network of positive ratings is sparse, small-world, and heavy-tailed without being strictly scale-free. It is disassortative, mostly because its hubs exceed the structural cutoff, and it is robust to random failures but fragile to targeted attacks. Its accumulated incoming-rating concentration rises from 2011 to 2015, including within

a fixed cohort, while the apparent post-2013 decline in yearly concentration is not supported at equal volume. The advantage of highly rated users depends strongly on the benchmark, falling from 12.70 under uniform choice to about 1.65 after matching tenure and recent activity, leaving a modest residual of unknown cause. These results do not establish causal preferential attachment or current commercial trust.

Data and code availability

The repository has the code, results, reproduction steps and data checksum [1].

Declaration on the use of AI tools

We used AI tools in this work: Codex (OpenAI) to draft analysis code, notebooks, figures and early text; Claude (Anthropic) to help write the degree-correlation script and notebook; and Claude Code (Anthropic) to audit the project against the brief, extend the theory comparison, integrate the degree-correlation results, revise the report, and build the Cytoscape figures, presentation and website. All reported numbers come from our own runs on the SNAP data. We checked the AI-assisted work by rerunning the full pipeline with the reproduction checker, executing all notebooks, recomputing formula predictions, comparing tables with the result files, and reviewing every figure and text. We take full responsibility for the content.

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